

Department of Civil Engineering, IIT Madras
CV 5101 (Modeling, Uncertainty and Data for Engineers)
Practice Questions (10 Oct, 2026)

1. **(Short Answer)** Answer the following brief questions:

- (a) What are the applications of the 3rd and 4th moments of a random variable?
- (b) When a random variable is the result of several additive effects, which distribution does it tend to follow? What distribution does it follow if the individual effects are multiplicative?
- (c) Provide two practical engineering examples where the Extreme Value Distribution is commonly applied.
- (d) In statistical hypothesis testing, which of the two error types (Type I or Type II) is usually considered worse in standard practice, and why?
- (e) Give a real-world engineering or decision-making example where committing a Type II error leaves us significantly worse off than committing a Type I error.

2. **(Uncertainty Propagation)** Let X_1, X_2 , and X_3 be mutually independent random variables with means μ_1, μ_2, μ_3 and variances $\sigma_1^2, \sigma_2^2, \sigma_3^2$, respectively. Consider a linear combination $Y = 2X_1 - 3X_2 + X_3 + 5$. Derive the exact expressions for the mean μ_Y and variance σ_Y^2 of Y .

3. **(Taylor Series Approximation)** A non-linear performance function of an engineering system is given by $Y = \ln(X) + \frac{2}{X} + e^{0.5X}$, where X is a random variable with mean $\mu_X = 4$ and variance $\sigma_X^2 = 0.64$. Compute:

- (a) The second-order Taylor series approximation for the expected value $E[Y]$.
- (b) The first-order Taylor series approximation for the variance $\text{Var}(Y)$.

4. **(Least-Squares Parameter Estimation)** Consider a non-linear structural response model given by $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 \ln(x)$. Given a dataset of n empirical observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$:

- (a) Formulate the design matrix $\mathbf{X}$ and the target vector $\mathbf{y}$.
- (b) Write down the complete matrix equation used to compute the least-squares parameter vector $\hat{\boldsymbol{\beta}} = [\beta_0, \beta_1, \beta_2, \beta_3]^T$.

5. **(Method of Moments Estimation)** Suppose the continuous probability density function (PDF) of a random variable X representing rainfall intensity is given by:

$$f(x; \theta) = \frac{2x}{\theta^2}, \quad 0 \leq x \leq \theta$$

Given a sample of size n with sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$:

- (a) Express the theoretical population mean $E[X]$ in terms of θ .
- (b) Derive the Method of Moments estimator $\hat{\theta}_{\text{MoM}}$ for the parameter θ .

6. **(Maximum Likelihood Estimation)** The time to failure T (in hours) of a hydraulic pump follows an exponential distribution with rate parameter $\lambda > 0$, having PDF $f(t; \lambda) = \lambda e^{-\lambda t}$ for $t \geq 0$.

- (a) Derive the log-likelihood function $\ln L(\lambda)$ for an observed random sample $t_1, t_2, \dots, t_n$.
- (b) Derive the Maximum Likelihood Estimator (MLE) $\hat{\lambda}_{MLE}$ for λ .

7. **(Confidence Interval for Population Mean)** A concrete manufacturing facility tests $n = 25$ cylinder specimens for compressive strength. The sample mean is calculated to be $\bar{x} = 32.5$ MPa. Assuming the compressive strength is normally distributed with a known population standard deviation $\sigma = 4.0$ MPa:

- (a) Construct a 95% confidence interval for the true mean compressive strength μ .
- (b) What sample size n would be required to narrow the total width of this confidence interval to 1.5 MPa at the same 95% confidence level? ($z_{0.025} = 1.96$).

8. **(Confidence Interval for Population Variance)** The variability in thickness of asphalt pavement layers must be tightly controlled. A sample of $n = 10$ pavement thickness measurements yields a sample variance $s^2 = 2.25$ mm². Assuming pavement thickness follows a normal distribution:

- (a) Construct a 90% confidence interval for the true population variance σ^2 . (Use critical values $\chi_{0.95,9}^2 = 3.325$ and $\chi_{0.05,9}^2 = 16.919$).
- (b) State the corresponding 90% confidence interval for the population standard deviation σ .

9. **(Central Limit Theorem & Standard Error)** Let $X_1, X_2, \dots, X_n$ be an independent and identically distributed (i.i.d.) random sample from an arbitrary population with mean μ and variance σ^2 .

- (a) Derive the mean and variance of the sample mean $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.
- (b) Explain how the Central Limit Theorem allows us to make probability statements about $\bar{X}_n$ even when the underlying population distribution is non-normal.

10. **(Hypothesis Testing on Mean - Known Variance)** A highway bridge deck is designed to withstand a maximum standard load with a mean deflection of no more than 12.0 mm. A field assessment team measures $n = 36$ deflection readings under peak load, obtaining a sample mean of $\bar{x} = 12.8$ mm. The population standard deviation is known to be $\sigma = 2.1$ mm. Conduct a hypothesis test at $\alpha = 0.05$ significance level to determine if the mean deflection exceeds design limits ($z_{0.05} = 1.645$). State H_0, H_1 , test statistic, decision rule, and conclusion.

11. **(Hypothesis Testing Concepts & Errors)** In environmental monitoring, a municipality tests water samples for lead concentration. The regulatory safety threshold is 15 ppb.

- (a) Formulate the null hypothesis H_0 and alternative hypothesis H_1 if the city wants to prove the water is unsafe before taking corrective action.
- (b) Describe the operational and public health consequences of making a Type I error in this context.

(c) Describe the consequences of making a Type II error in this context.

12. **(Hypothesis Testing - Power & Type II Error)** An automated sorting mechanism in a structural steel fabrication yard checks soil compaction density. The base compaction level is $\mu_0 = 95\%$, with known standard deviation $\sigma = 3\%$. A sample of $n = 25$ measurements is evaluated at significance level $\alpha = 0.05$ for a one-tailed test ($H_0 : \mu = 95\%$ vs $H_1 : \mu > 95\%$).

(a) Calculate the critical sample mean value $\bar{x}_{\text{crit}}$ above which H_0 is rejected ($z_{0.05} = 1.645$).

(b) If the true mean compaction level is actually $\mu_1 = 97.5\%$, calculate the probability of committing a Type II error (β). ($z_{0.025} = 1.96, z_{0.05} = 1.645$).

13. **(Probability & Covariance Properties)** For two continuous random variables X and Y , prove that $\text{Cov}(aX+b, cY+d) = ac \text{Cov}(X, Y)$, where a, b, c, d are deterministic constants.
